Knotweed Checker

Spot invasive Japanese knotweed from a phone photo

AI for Good, Hackathon 7: Eyes on Earth · Roboflow · SDG 15

The problem

- Japanese knotweed pushes out native plants, can damage river banks and paving, and regrows from tiny root pieces
- Amsterdam set aside €8.2 million to fight it; Ameland and Heusden ask residents to report where it grows
- Patches are cheapest to remove when found early
- But most people cannot tell knotweed from look-alikes such as bindweed



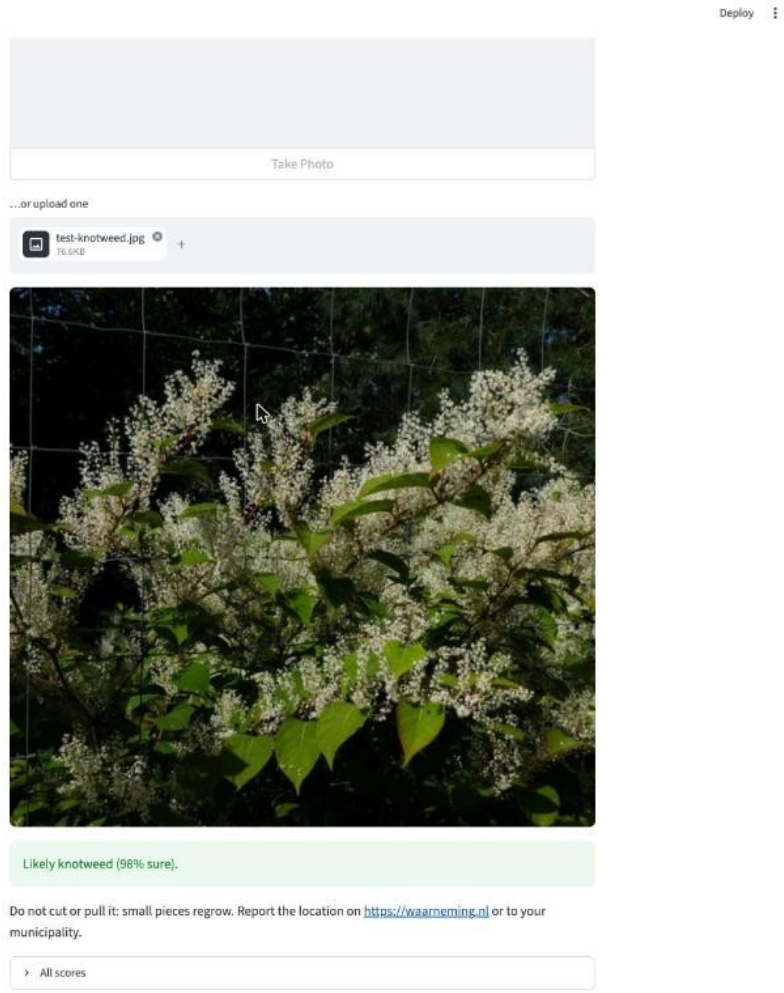
Source: gemeente.nu, 2019, 'Strijdplan Japanse duizendknoop samen met inwoners'. SDG 15, target 15.8: invasive alien species.

Who it is for (and who not)

- For: residents and park volunteers in the Netherlands who see a suspicious plant and want to know in seconds whether to report it
- They have a phone, no botany knowledge, and little time
- Not for: municipal staff or contractors deciding whether to remove a plant; that needs an expert on site
- Not for: identifying other species

How it works

- 1. Take or upload a photo in the app
- 2. The app sends it to our Roboflow model: the step that decides 'knotweed or not'
- 3. 70%+ → 'Likely knotweed: don't cut it, report it on waarneming.nl'
- below 70% → 'Maybe: take a closer photo' · otherwise 'Probably not'
- Why not a general AI model? We tested one (CLIP): same knotweed found, but 4x more false alarms



Data and model

- 832 Dutch iNaturalist photos, community-confirmed species, Creative Commons licences (credits in sources.csv)
- 302 knotweed · 530 look-alikes: bindweed, Persicaria, Bistorta, Fallopia, bramble, nettle, ivy
- Bug we caught: iNaturalist's search mixed knotweed and bindweed, so ~70 photos had the wrong label
- Split by plant, not by photo (587 / 172 / 73), so the test never shows a plant the model already saw
- Roboflow classification, ViT model, training photos doubled with flips, rotations and brightness changes

Results on photos it never saw

- 73 Dutch test photos: 95% right · 24 of 26 knotweed found · 2 of 47 false alarms
- 168 new photos from Belgium and Germany, all seasons: 93% right · 77 of 83 found · 5 of 85 false alarms
- Spring 93% · summer 93% · autumn/winter 95%: spring shoots are only slightly harder
- General AI model (CLIP), same 168 photos: 84% · 77 of 83 found · 21 of 85 false alarms
- Our model: 4x fewer false alarms than the general model, which could mean fewer unnecessary checks
- Worst mistake: a bindweed called knotweed with 98% confidence

Ethics: what could go wrong, and what we did

- Missed knotweed → patch keeps spreading (the expensive mistake)
- False alarm → someone cuts or poisons a harmless plant; cutting real knotweed spreads it
- What we did: 70% threshold with a 'maybe' zone; the app never tells you to remove anything, only to report it; tested on a second, independent set
- Every answer warns it can be wrong; we report missed knotweed, not just accuracy
- Photos go to Roboflow's servers, so don't photograph people or house numbers
- Next: test on our own street photos; add winter images

Live demo and what we learned

- Demo: upload a photo → answer in a few seconds
- Learned: the labels you scrape can be wrong; check them before training
- Learned: a high score can hide the mistakes that matter; count the misses and false alarms
- Learned: our guess that spring would be much worse was wrong; testing beat guessing
- What surprised us: OUR_OWN_LESSON